LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Feedback, types and its effect on control system.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Feedback, types and its effect on control system.
5. Specific Objectives	20.1.1.1 Trainee will be able to classify control systems, feedback and its effect.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Diagramming of open and closed loop systems; Safety: Ensure diagrams are clear; Supervision: Instructor verifies loop identification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) Define a control system.
(b) What is the difference between open and closed loop systems?
(c) Why is feedback necessary in an aircraft control system?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Feedback • Open Loop Control Systems • Closed Loop Control Systems • Positive Feedback Mechanism • Negative Feedback Mechanism	L & Ds	PPT, IB	15 min
(b)	• Effect of Negative Feedback • Effect of Positive Feedback • Stability Analysis • Oscillation Risks • System Response Improvement	L & Ds	PPT, PDF	20 min
(c)	• Comparison of Loop Types • Real-world Aircraft Examples • Summary of Effects • Key Terminology Review • Preparation for Next Lesson	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how negative feedback improves system stability.
(b) Describe the effect of positive feedback on system oscillation.
(c) Identify one aircraft system that uses closed-loop control.

12. Home Assignment and Follow Up	Draw a block diagram of a closed-loop control system and label the feedback path.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Purpose of test signals used in control systems, Step function, Impulse function, RAMP functions, Parabolic functions.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min